Standard Model of Elementary Particles

	three generations of matter (fermions)			interactions / force carriers (bosons)	
	I	II	III		
mass charge spin					
QUARKS	≈$2.16\ \mathrm{MeV}/c^2$ ⅔ ½ u **up**	≈$1.273\ \mathrm{GeV}/c^2$ ⅔ ½ c **charm**	≈$172.57\ \mathrm{GeV}/c^2$ ⅔ ½ t **top**	0 0 1 g **gluon**	≈$125.2\ \mathrm{GeV}/c^2$ 0 0 H **higgs**
QUARKS	≈$4.7\ \mathrm{MeV}/c^2$ −⅓ ½ d **down**	≈$93.5\ \mathrm{MeV}/c^2$ −⅓ ½ s **strange**	≈$4.183\ \mathrm{GeV}/c^2$ −⅓ ½ b **bottom**	0 0 1 γ **photon**	SCALAR BOSONS
LEPTONS	≈$0.511\ \mathrm{MeV}/c^2$ −1 ½ e **electron**	≈$105.66\ \mathrm{MeV}/c^2$ −1 ½ μ **muon**	≈$1.77693\ \mathrm{GeV}/c^2$ −1 ½ τ **tau**	≈$91.188\ \mathrm{GeV}/c^2$ 0 1 Z **Z boson**	
LEPTONS	<$0.8\ \mathrm{eV}/c^2$ 0 ½ ν_e **electron neutrino**	<$0.17\ \mathrm{MeV}/c^2$ 0 ½ ν_μ **muon neutrino**	<$18.2\ \mathrm{MeV}/c^2$ 0 ½ ν_τ **tau neutrino**	≈$80.3692\ \mathrm{GeV}/c^2$ ±1 1 W **W boson**	GAUGE BOSONS VECTOR BOSONS